



Birthday square

You can adapt the magic square below so that the numbers add to any number bigger than 22. The secret is to change just the four highlighted numbers. At the moment, the magic sum is 22. Suppose you want to change it to 30. Because 30 is 8 more than 22, just add 8 to the highlighted numbers and draw out the square again. It always works!

8	11	2	1
1	2	7	12
3	4	9	6
10	5	4	3

Use this magic square to make a birthday card, with the numbers adding up to the person's age.

Upside-down square

96	11	89	68
88	69	91	16
61	86	18	99
19	98	66	81

See if you can work out the magic sum for this very unusual square. Then turn the page upside down and look at the square again. Does it still work?



A KNIGHT'S TOUR

In the magic square below, the rows and columns add up to 260. But there's something even more surprising about this square. Look at the pattern the numbers make as you count from 1 upwards. Each move is like the move of a knight on a chessboard: two steps forwards and one step to the side.

1	48	31	50	33	16	63	18
30	51	46	3	62	19	14	35
47	2	49	32	15	34	17	64
52	29	4	45	20	61	36	13
5	44	25	56	9	40	21	60
28	53	8	41	24	57	12	37
43	6	55	26	39	10	59	22
54	27	42	7	58	23	38	11

Make your own

magic square by using knight's moves. Draw a 5x5 grid and put a 1 anywhere in the bottom row. Fill in higher numbers by making knight's moves up and right. If you leave the grid, re-enter on the opposite side. If you can't make a knight's move, jump two squares to the right instead.

